

Solution Requirements

Date: 30 September 2026

Team ID: SWTID-2026-4040

Project Name: ComicCraft — AI Comic Story Creator Using Gemini Models

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Team Members: Sandhiya D, Rithikaa B, Anitha R, Anbuselvi

Team Size: 5

Maximum Marks: 4

Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Input	Accept story prompt, character name, setting, tone and art style through the web form.	High
2	Outline Generation	Generate a five-panel comic outline using Gemini Flash.	High
3	Story Generation	Expand the outline into narration and dialogue using Gemini Pro.	High
4	Image Generation	Generate panel artwork from image prompts using Stable Diffusion.	High
5	Layout	Combine panel metadata, images and story into a comic layout.	High
6	Preview	Display generated panels in a browser-based preview.	High
7	Export Flow	Provide a PDF export workflow and success page.	Medium
8	Error Handling	Handle invalid input, missing credentials and service/runtime failures gracefully.	High

Non-Functional Requirements (NFR)

- Usability: the workflow should be understandable to a first-time user.
- Maintainability: separate modules for routes, Gemini integrations, image generation and layout.
- Security: API credentials must be supplied through environment configuration rather than hard-coded source.
- Portability: run in a compatible Python environment with the listed dependencies.
- Performance: AI calls and local image generation may be compute-intensive; response targets should be measured in deployment.
- Reliability: failures from external model services should not cause an unhandled user-facing crash.